

SEDAR 24: South Atlantic Red Snapper

Data Workshop

Workshop Term of Reference #10 called for the Data Panel to provide recommendations for future research in areas such as sampling, fishery monitoring, and stock assessment; and to include specific guidance on sampling intensity (number of samples including age and length structures) and appropriate strata and coverage.

Life History Work Group

The life history WG recommended the following:

- **Age Reading Comparisons** Continuing the age reading comparisons and calibrations between labs on a reference collection of known age fish would be beneficial for determining a more accurate aging error matrix and would provide accuracy to the age composition data.
- **Movements and Migrations** More research on red snapper movements/migrations in Atlantic waters is needed. Available data and the results of studies in the Gulf of Mexico indicate high site fidelity. Tropical storms may cause greater than normal movement.

Commercial Work Group

The Workgroup reviewed recommendations from SEDAR 15 and offers additional recommendations. The Commercial WG notes that Sea Grant is currently funding a video monitoring program for observing the snapper-grouper fishery using exemption permits with 7 total vessels participating (1 in NC, 2 each in SC, GA, and FL).

The commercial WG recommended the following:

- **Electronic Logbooks**
- **More observers**
 - 5-10% allocated by strata within states
 - Possible to use exemption to bring in everything with no sale
 - Get maximum information from fish
- **Angler education with regards to recording depths on paper logbooks**
- **More precise depths by species from port agents (would require data base change)**
- **Expand TIP sampling**
 - Reallocate samplers for at-sea observer trips
 - Improve sampling from Florida's handline and dive gear where most of the effort and landings are from.
- **Continue to sample more ages (proportional to effort), although large numbers of ages were sampled in the most recent years, especially 2009.**

Recreational Work Group

The recreational WG recommended the following:

- In order to separate PR and CH catch data, more age data are needed, particularly from the PR mode.
- Continued research efforts to incorporate/require logbook reporting from recreational anglers.
- Quantify historical fishing photos for use in future SEDARS.
- MRFSS At-SEA observer program in NC, SC and GA should collect depth fished data. Standardize data elements within this program.
- Headboat Survey logbook should also collect depth information.
- Continued research efforts to collect discard length and age data from the private sector.
- Improve metadata collection in the recreational fishery.

Indices Work Group

The indices Work Group recommended the following:

- More fishery independent data collection
- Exploration of the Stephens and MacCall trip selection method and alternatives methods
 - Explore the use of actual landings rather than presence/absence for other species for trip selection
- Evaluate how fishermen preferences change over time and whether such changes affect CPUE
- Increase observer coverage, including information on area fished and depth
- Examine how catchability has changed over time with increases in technology and potential changes in fishing practices. This is of particular importance when considering fishery dependent indices
- Investigate potential density-dependent changes in catchability

Analytic Approach Work Group

There were no research recommendations from the Analytic Approach working group.

Discards Mortality Work Group

The discards mortality WG recommended the following:

- More hooking, size, and depth related discard mortality studies
- Angler education
- More accurate depths by species from logbooks
- Survey of fishermen and scientists to possibly get information on depth of areas fished and species abundance

More species specific depth information collected by port agents

Assessment Workshop

Assessment Process I Term of Reference #10 called for the Assessment Panel to provide recommendations for future research and data collection.

The assessment panel recommended the following:

- Fishery independent surveys of reef fishes in the Southeast were expanded in 2010 and continued expansion is recommended. These data should be made available for future assessments of red snapper.
- More information on age/length composition of discards from various fleets would improve stock assessment of reef fishes in the Southeast, including red snapper. A recreational discard reporting system would benefit future assessments.
- More information on discard mortality rates would improve stock assessment of reef fishes in the Southeast, including red snapper.
- More detailed spatial and temporal resolution of fishing effort for each fleet would likely improve assessments.
- Methods to characterize uncertainty in assessment results deserve further consideration. For avoiding overfishing, characterizing uncertainty is more than an academic exercise, particularly when relying on probabilistic methods to set catch levels.
- Compared to other fishes, red snapper mature very young relative to their life span. This is consistent with the hypothesis that maturation, as a character trait, is influenced by exploitation. Assessments and management could be improved by better understanding plasticity in life-history traits (such as maturation), as well as evolutionary effects of exploitation.
- Depth appears to be important component of population and fishery dynamics for red snapper. Spatial assessment models might be able to address depth explicitly, if migration rates among strata were better understood.
- Increased fishery independent sampling of larvae and juveniles.
- Increased TIP sampling.
- Increased sampling of recreational sector.
- Examine or develop ways to include anecdotal information in SEDAR assessments.

Review Workshop

The Review Panel suggested some recommendations, categorized as more important (Tier 1) and less important (Tier 2).

Tier 1

- Investigate alternate stock recruitment models, and in particular the robustness of stock status conclusions to reasonable alternative stock-recruit assumptions.

- Consider estimating missing catch (e.g., recreational) within the model to improve consistency. An example of such an approach is the B-ADAPT model applied to North Sea cod.
- Review historical records for determining historical average weights of fish. This is consistent with a DW recommendation.
- The Review Panel agreed with the DW and AW recommendations to improve age sampling. In particular, this should improve the estimation of fishing mortality in BAM.
- The Review Panel agreed with the DW and AW recommendations to continue developing fishery-independent abundance indices, especially because assumed changes in catchability of CPUE indices for red snapper are uncertain.
- Explore changes in catchability in light of other species involved in the mixed species fisheries that catch red snapper. The Review Panel anticipates that changes in catchability may be consistent among some of these species.

Tier 2

- Consistent with the AW recommendation regarding “plasticity in life-history traits”, the Review Panel recommends investigating for temporal variation in growth and maturation rates, especially when such characteristics often show a density-dependent response.
- Tagging studies can provide relatively direct estimates of fishing mortality and selectivity, growth rates, and other stock assessment parameters. Where possible, information from tagging studies that are representative of the stock as a whole should be incorporated into the assessment.